

8086 Instruction Set Architecture

Q. Classify the Instruction Set of the 8086 microprocessor. Briefly explain Data Transfer, Arithmetic, Logical, Branch, String, and Processor Control instructions with one example each.

[DEF] Definition to Remember

An Instruction Set is the complete collection of commands a microprocessor can understand and execute. The 8086 Instruction Set tells the CPU how to manipulate data in registers/memory, perform math, and control the flow of execution.

1. Classification of Instructions

A. Data Transfer Instructions

Moves

B. Arithmetic Instructions

Perform

C. Logical Instructions

Perform

D. Branch / Control Transfer Instructions

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E. String Instructions

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F. Processor Control Instructions

Contro

[MUST-WRITE] Must-Write Points (for 10 marks)

- 1. An Instruction Set defines all commands the 8086 CPU can execute.
- 2. Data Transfer: Moves data (e.g., MOV AX, BX). Does not alter flags.
- 3. Arithmetic: Math operations (e.g., ADD AL, 05H). Updates flags.
- 4. Logical: Bitwise operations (e.g., AND CX, DX, SHL).
- 5. Branch/Control: Alters execution flow (e.g., JMP, JZ). Changes the IP register.
- 6. String: Operates on blocks of memory using SI and DI (e.g., MOVSB).
- 7. Processor Control: Manages hardware state/flags (e.g., STC, HLT).

[RECALL] Quick Recall

8086 Instructions -> Data (MOV) -> Arithmetic (ADD/INC) -> Logical (AND/SHL) -> Branch (JMP/JZ) -> String (MOVSB) -> Control

(STC/HLT)